LOT Pager — Software that Connects to the Firmware

Document: LOT-HW-01 / SW · **Rev:** A (2026-11-13)

Three pieces of software talk to the firmware. None needs a phone app.

1. Setup page (web, inside lot-systems.com)

- Route: `/settings/devices/setup` in the existing React client.
- Uses **Web Bluetooth** (Chrome/Edge desktop + Android) to reach the device in `UNPAIRED` state.
- Flow:
  1. User signs in → "Add LOT Pager" → server issues pair code (doc 05 §3).
  2. Browser connects over BLE (service UUID `LOT-PAGER-SETUP`), sends Wi-Fi SSID + password + pair code.
  3. Device joins Wi-Fi, calls `pair/finish`, stores token, shows `PAIRED ✓`.
- iPhone Safari has no Web Bluetooth → fallback: device shows a pair code and starts a temporary Wi-Fi access point `LOT-XXXX` with a captive setup page (ESP-IDF SoftAP provisioning).

2. Factory / service tool (`hardware/tools/lot-flash`)

- Node.js CLI + Web Serial page; talks to the pogo-pin UART.
- Commands: `flash` (esptool, factory + app images), `test` (runs factory self-test, parses JSON result), `provision` (reads ATECC public key and serial, registers the device with `/admin-api/devices/register`), `label` (prints serial label).
- Writes one CSV line per unit → manufacturing record (doc 08).

3. Device simulator (`scripts/hardware/device-sim.ts`)

- Acts like a paired device against a local or staging server: polls inbox, prints notices, sends `copy` on Enter, sends fake BME688 sessions (CBOR + heatshrink).
- Lets the full loop (AI notice → device → Copy → Log tab) be developed and tested before any PCB exists. Built in session 2 together with the API.

4. OTA release pipeline

- GitHub Action on tag `pager-fw-vX.Y.Z`: build with ESP-IDF Docker image → sign → upload to object storage → update manifest served by `GET /api/device/ota`.
- Staged rollout: 10 % of devices, then 100 % after 48 h without crash reports.

5. Interfaces summary

Link	Transport	Format
Device ↔ lot-systems.com	HTTPS (TLS 1.2+)	JSON (small calls), CBOR + heatshrink (sessions), JPEG
Browser ↔ device (setup)	BLE GATT or SoftAP HTTP	protobuf (`wifi_provisioning`)
Factory PC ↔ device	UART 115200 via pogo pins	esptool + JSON lines